2025 AMC 10B Problem 10 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 10. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 10

Problem:

Let $f(n) = n^3 - 5n^2 + 2n + 8$, and let $g(n) = n^3 - 6n^2 + 5n + 12$. What is the sum of all integer values of n for which $\dfrac{f(n)}{g(n)}$ is also an integer?

Answer Choices:

A. 2

B. 3

C. 4

D. 5

E. 6

Solution:

Let

$$f(n) = n^3 - 5n^2 + 2n + 8 \qquad g(n) = n^3 - 6n^2 + 5n + 12.$$

Using the Remainder Theorem,

$$\begin{aligned} f(-1) = f(2) = f(4) = 0 \ &\Rightarrow\ f(n) = (n+1)(n-2)(n-4), \\ g(-1) = g(3) = g(4) = 0 \ &\Rightarrow\ g(n) = (n+1)(n-3)(n-4). \end{aligned}$$

Hence,

$$\frac{f(n)}{g(n)} = \frac{(n+1)(n-2)(n-4)}{(n+1)(n-3)(n-4)} = \frac{n-2}{n-3},$$

with the restrictions

$$n \neq -1,\ 3,\ 4$$

(since these make the denominator of the original fraction zero). Now,

$$\frac{f(n)}{g(n)} = \frac{n-2}{n-3} = \frac{(n-3)+1}{n-3} = 1 + \frac{1}{n-3}.$$

For $\dfrac{f(n)}{g(n)}$ to be an integer, we must have

$$\frac{1}{n-3} \in \mathbb{Z}.$$

Thus $n-3$ must divide 1, so

$$n - 3 = \pm 1 \quad \Rightarrow \quad n = 4 \text{ or } n = 2.$$

But $n = 4$ is not allowed (it makes $g(4) = 0$), so the only integer value of n satisfying the condition is $n = 2$. Therefore, the sum of all integer values of n for the given condition is $\boxed{\textbf{(A) } 2}$.